PhonePe Pulse Data Visualization & Exploration

- **A Deep Dive into India's Digital Payment Ecosystem (2018-2024)**

- **Tech Stack:** Python, Streamlit, Plotly, MySQL
- **Presented by:** Kartik Bingi

Project Objective

•**The Goal:** To transform raw, complex JSON data from the PhonePe Pulse GitHub

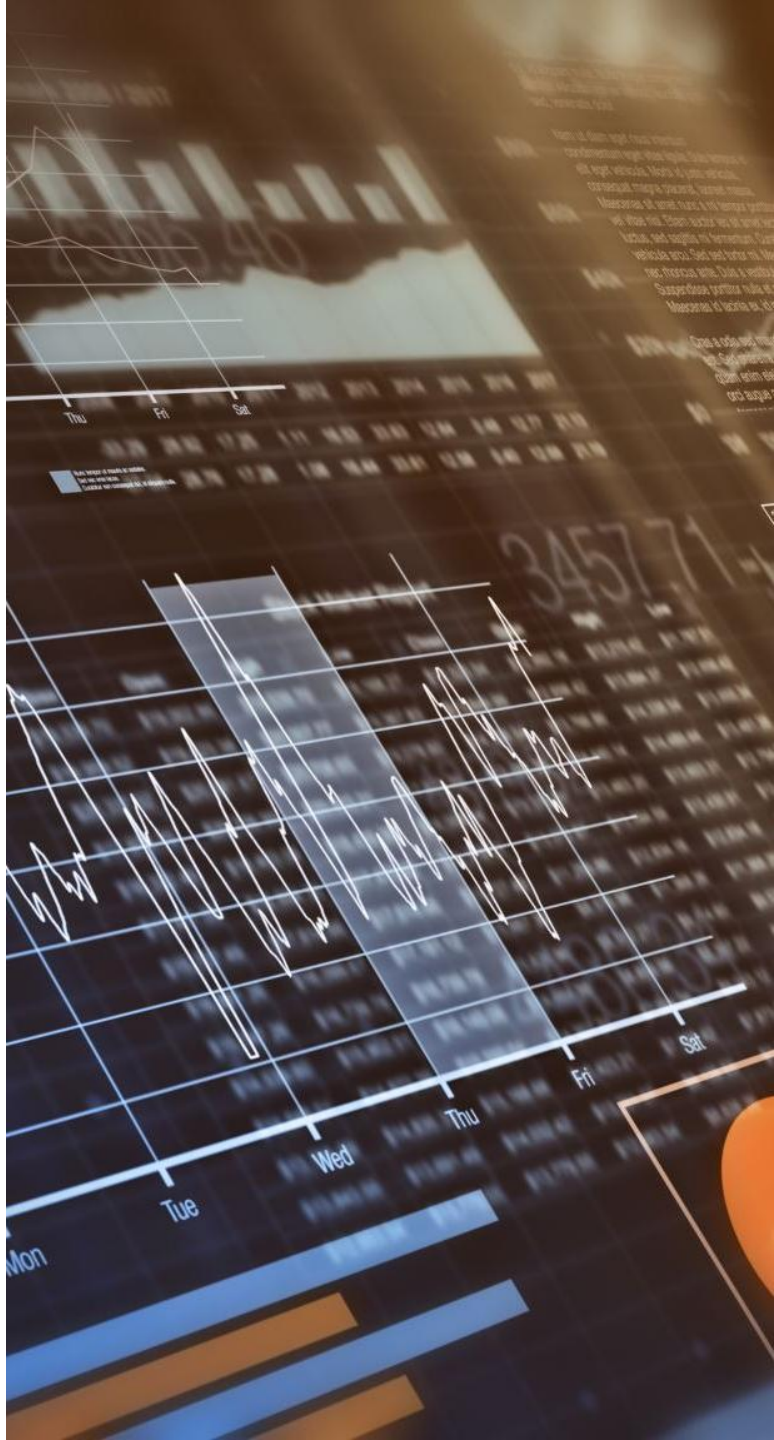
repository into an interactive, visual dashboard.

•**Key Focus Areas:**

•State-wise transaction growth.

•User demographics and device preferences.

•Identification of top-performing districts and categories.



Data Pipeline (ETL Process)

1

Extract: Cloning the PhonePe Pulse GitHub repo and extracting nested JSON files.

2

Transform: Using Python and Pandas to clean data, handle missing values, and convert "kebab-case" names to "Title Case."

3

Load: Storing structured data in a MySQL Database (local) and CSV files (cloud) for high-speed retrieval.

Dashboard Architecture

Frontend: Streamlit for the user interface.

Visuals: Plotly Express for dynamic Choropleth maps and Sunburst/Donut charts.

Deployment: Streamlit Cloud for global accessibility.

Key Feature: Cached data loading using for a lag-free user experience.

Key Insights & Visualizations

Geographic Heatmap: Visualizes which states are leading in digital adoption.

Transaction Categories: Breakdown showing the dominance of P2P (Peer-to-Peer) vs.Merchant payments.

Growth Trends: How the "Avg. Ticket Size" has evolved post-2021.

Challenges & Solutions

CHALLENGE: LARGE, NESTED JSON STRUCTURES WERE DIFFICULT TO QUERY.

SOLUTION: DEVELOPED A RECURSIVE PYTHON SCRIPT TO FLATTEN DATA INTO A RELATIONAL FORMAT.

CHALLENGE: MAPPING GEOJSON TO CUSTOM STATE NAMES.

SOLUTION: APPLIED STRING MANIPULATION TO SYNC CSV DATA WITH INDIA'S OFFICIAL GEOGRAPHIC LABELS.

Conclusion & Future Scope

Conclusion: The dashboard provides a 360-degree view of PhonePe’s market presence.

Future Scope: * Integrating live API data for real-time updates.

Adding Predictive Analytics to forecast next year's transaction volume.

THANK YOU